

Report: Cardio Risk Prediction

1. Exploratory Data Analysis (EDA)

❖ Data Cleaning & Preprocessing:

- **ID Removal & Type Casting:** Dropped uninformative columns (id, Unnamed: 0). Converted categorical columns (cholesterol, gluc, smoke, alco, active, gender) from integers to proper categorical data types.
- **Unit Standardization:** Converted age from days to years and height from centimeters to meters.
- **Outlier Filtering:** Filtered impossible values using clinical limits: blood pressure outside 30–300 mmHg, cases where “ap_lo” >= “ap_hi”, “height” <100 cm or >250 cm, and “weight” <30 kg.

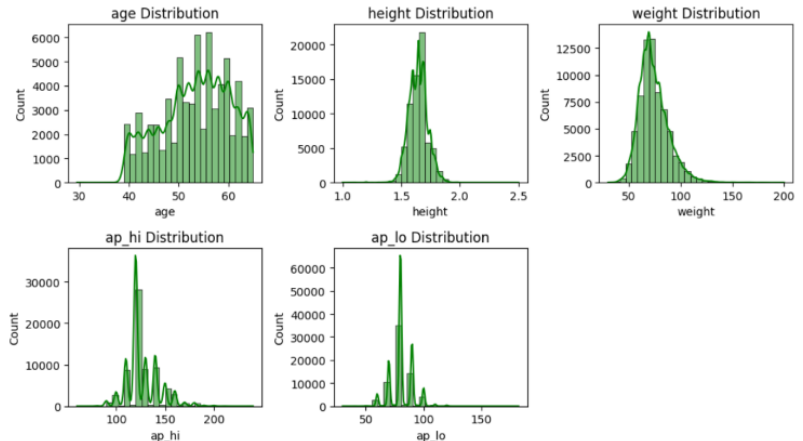
❖ Feature Distributions & Univariate Analysis:

- **Demographics & Physicals:** Most patients are aged 40-65. Gender exhibits an imbalance (44,715 in Category 0 vs. 23,931 in Category 1). Height and weight show normal distributions, while blood pressure features spike at standard rounded readings (e.g., 120/80 mmHg).
- **Health & Habits:** Normal levels dominate cholesterol (51,477) and glucose (58,363). Smoking (6,037) and alcohol use (3,662) are low, while activity (55,148) is high.
- **Class Balance:** Target (cardio) is balanced (34,684 class 0 vs. 33,962 class 1), requiring no synthetic resampling.

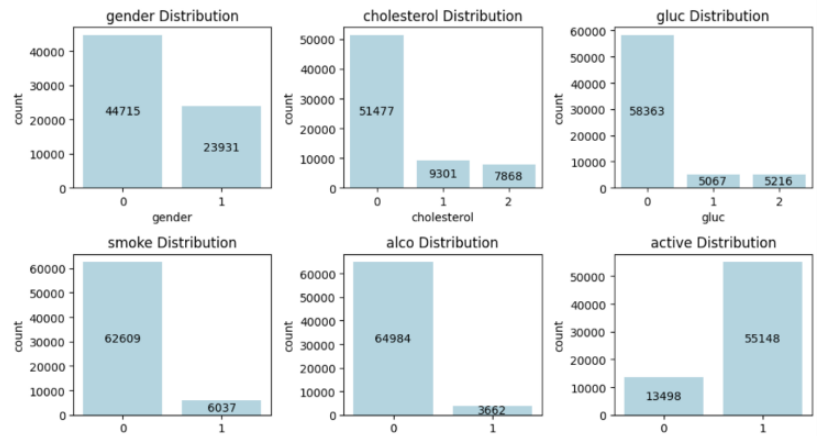
❖ Bivariate Analysis & Key Insights:

- **Multicollinearity:** Strong linear correlation ($r=0.73$) exists between “ap_hi” and “ap_lo”.
- **Risk Proportion:** Target likelihood increases with severity in cholesterol (43.5% >> 59.6% >> 76.3% for levels 0, 1, 2) and glucose (47.6% >> 58.9% >> 61.8% for levels 0, 1, 2).

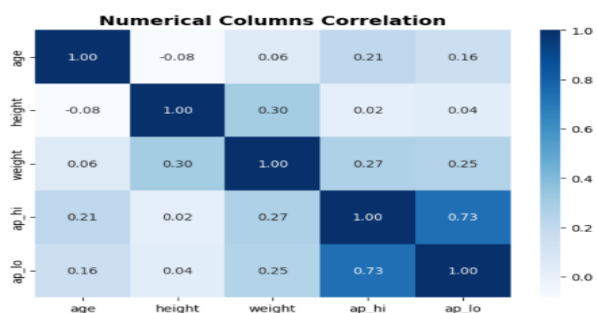
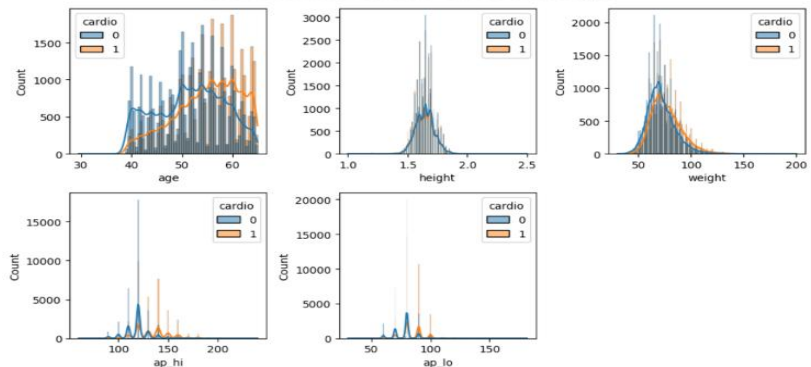
Data Distribution Analysis of Numerical Features



Categorical Column Distribution



Plot Distribution for each feature with target



- **Primary Drivers:** Age and blood pressure are the strongest predictors of risk, weight is a moderate driver, while height shows zero impact on target separation.

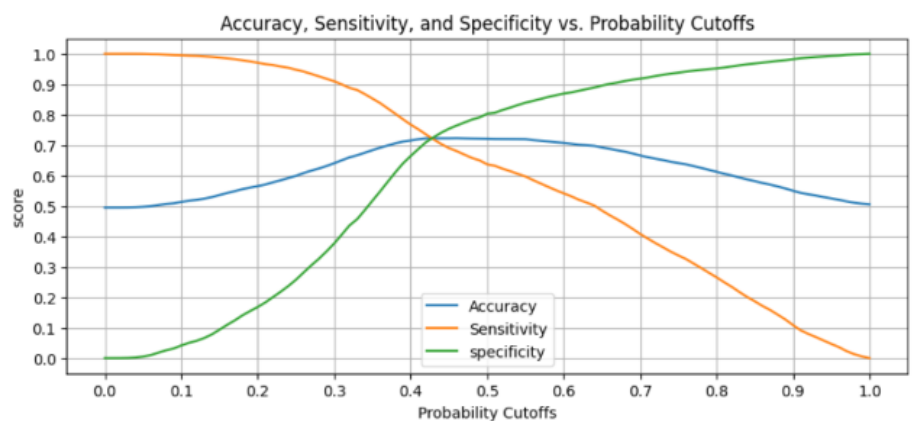
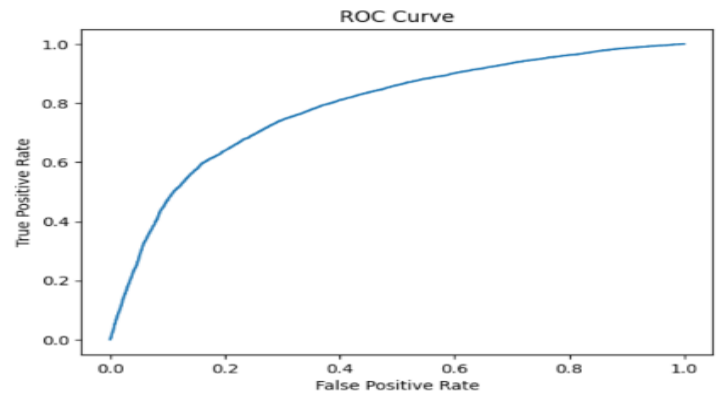
2. Model Building and Rationale

❖ Feature Engineering & Selection:

- **Weight Dropped:** Engineered BMI ($\text{weight} / \text{height}^2$) and dropped weight due to high correlation ($r=0.85$) to eliminate multicollinearity while preserving BMI. Retained both “ap_hi” and “ap_lo” despite correlation ($r = 0.73$) as both are essential clinical indicators.
- **Glucose Regrouping:** Combined low-frequency “gluc” levels (1 and 2) into a single elevated category (1) representing ~15% of data. Retained all cholesterol categories due to significant representation in higher levels (~25% combined).
- **Encoding & Scaling:** One-hot encoded cholesterol and scaled features using MinMaxScaler.

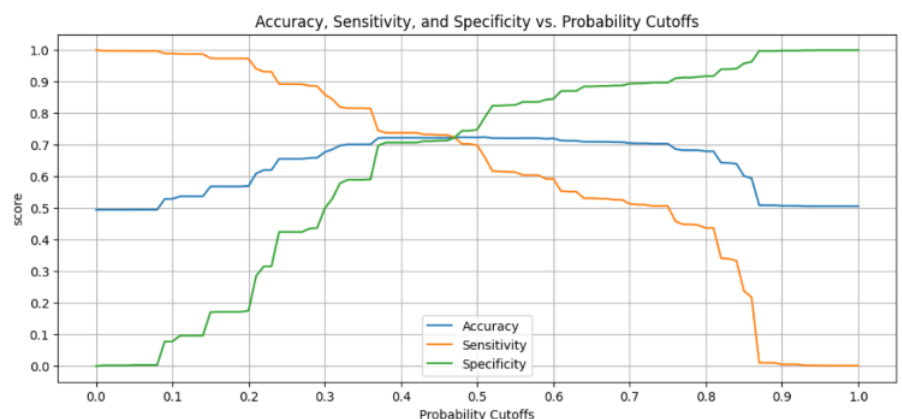
❖ Threshold Optimization:

- **Probability Cutoffs:** Evaluated `predict_proba()` vs default `.predict()` on Linear SVC. Probability predictions improved recall for CVD cases (0.65 vs 0.62), which is critical for medical screening.
- **Optimal Cutoffs:** Selected probability thresholds where sensitivity and specificity curves intersect to maximize overall balance:
- **Linear SVC:** Cutoff = 0.42 (sensitivity/specificity ~72%).
- **Tuned Decision Tree:** Cutoff = 0.465 (sensitivity/specificity ~73%).



❖ Overfitting & Hyperparameter Tuning:

- **Baseline Tree Overfitting:** The initial decision tree scored 100% across all metrics on training data, showing extreme overfitting.
- **GridSearchCV Tuning:** Optimized hyperparameter grid using 5-fold CV. Best parameters: `criterion= 'gini'`, `max_depth= 7`, `min_samples_leaf= 1`, `min_samples_split= 2`.
- **Result:** Constraining depth to 7 eliminated overfitting, matching train (74%) and test (72%) performance.



❖ Model Comparison & Conclusion

- **Linear SVC (Train / Test):**
 - Accuracy: 0.73 / 0.72
 - Precision (Class 1): 0.72 / 0.71
 - Recall (Class 1): 0.75 / 0.73

- F1-Score (Class 1): 0.73 / 0.72
- **Tuned Decision Tree (Train / Test):**
 - Accuracy: 0.74 / 0.72
 - Precision (Class 1): 0.73 / 0.72
 - Recall (Class 1): 0.75 / 0.72
 - F1-Score (Class 1): 0.74 / 0.72
- **Final Selection:** Both models achieve 72% test accuracy. Linear SVC is selected as the primary model due to higher test recall for CVD patients (0.73 vs 0.72), maximizing early risk detection.

3. Model Cards

Model Card 1: Linear Support Vector Classifier (SVC)

❖ Model Overview:

- This Linear Support Vector Classifier (kernel='linear', probability=True) was trained on 48,052 patient records to classify binary cardiovascular disease status (cardio). Its primary role is to serve as a preliminary risk screening tool in clinical settings to assist medical staff during routine patient checkups.

❖ Data and Preprocessing Steps:

- **Input features used:** age, gender, height, ap_hi, ap_lo, cholesterol, gluc, smoke, alco, and active.
- **Feature Engineering:** Calculated BMI (weight / height^2) and subsequently dropped weight to eliminate strong multicollinearity ($r = 0.85$). Retained both ap_hi and ap_lo due to their independent clinical value despite a $r = 0.73$ correlation.
- **Categorical Transformations:** Grouped low-frequency gluc categories (1 and 2) into a single elevated flag (~15% of the split). One-hot encoded cholesterol into separate dummy columns. All numerical variables were scaled using MinMaxScaler.

Model Setup and Execution:

- **Algorithm:** Support Vector Classifier with linear kernel.
- **Configuration:** probability=True enabled to retrieve probability scores; random_state=42.
- **Threshold Choice:** Lowered the decision threshold from 0.50 down to 0.42 after evaluating the sensitivity-specificity intersection plot (~0.72 score).

Performance Breakdown:

- **Training Data (Cutoff = 0.42):** 0.73 Accuracy | 0.72 Precision | 0.75 Recall | 0.73 F1-Score
- **Test Data (Cutoff = 0.42):** 0.72 Accuracy | 0.71 Precision | 0.73 Recall | 0.72 F1-Score
- **Practical Strengths:** Outperformed both the default 0.5 threshold (0.65 recall) and the tuned decision tree (0.72 recall) by raising test recall for CVD patients to 0.73.
- **Practical Trade-offs:** Precision sits at 0.71 on test data, which resulted in 3,017 false positive flags out of 20,594 test records.

Real-World Limitations & Evaluation Details:

- **Logic Constraints:** Model behavior is determined by weights across a linear hyperplane over 12 scaled inputs, meaning it doesn't offer straightforward if-then branching rules.
- **Misclassification Risk:** At our 0.42 cutoff, 2,751 true positive CVD cases in the test set were wrongly predicted as negative (False Negatives).

- **Dataset Nuances:** The raw dataset contains a noticeable gender distribution split (65% Category 0 vs 35% Category 1).
- **Data Quality Dependency:** Outputs depend heavily on accurate blood pressure inputs (ap_hi and ap_lo) without severe measurement errors.

Model Card 2: Tuned Decision Tree Classifier

❖ Model Overview:

- This Decision Tree model was built on 48,052 training samples and tuned via 5-fold cross-validation. It offers a transparent, rule-based approach for healthcare teams who require clear decision splits when predicting cardiovascular risk.

❖ Data and Preprocessing Steps:

- **Features used:** age, gender, height, ap_hi, ap_lo, cholesterol, gluc, smoke, alco, active, and BMI.
- **Transformation logic mirrors the SVC pipeline:** weight dropped due to collinearity with BMI ($r = 0.85$), gluc levels 1 and 2 merged into one category, cholesterol dummy-encoded, and values normalized via MinMaxScaler.

❖ Model Setup and Execution:

- **Algorithm:** Decision Tree Classifier.
- **Hyperparameter Tuning:** Ran a 5-fold GridSearchCV optimizing for f1. Best parameters: criterion='gini', max_depth=7, min_samples_leaf=1, min_samples_split=2.
- **Fixing Overfit:** The raw baseline tree scored 100% across all metrics on training data. Restricting max_depth to 7 fixed this completely.
- **Threshold Choice:** Set optimal decision threshold to 0.465 based on where sensitivity and specificity curves meet.

❖ Performance Breakdown:

- **Training Data (Cutoff = 0.465):** 0.74 Accuracy | 0.73 Precision | 0.75 Recall | 0.74 F1-Score
- **Test Data (Cutoff = 0.465):** 0.72 Accuracy | 0.72 Precision | 0.72 Recall | 0.72 F1-Score
- **Feature Importance Highlights:** The top drivers guiding tree splits were age (0.282), ap_hi (0.231), and BMI (0.216).
- **Practical Trade-offs:** Slightly lower test recall for CVD patients (0.72) compared to the Linear SVC model (0.73).

❖ Real-World Limitations & Evaluation Details:

- **Depth Constraints:** Feature evaluation stops at a maximum tree depth of 7 across the 12 features.
- **Misclassification Risk:** With a 0.72 test recall at the 0.465 cutoff, the model misses 28% of actual CVD cases.
- **Heavy Feature Dependency:** Splitting logic relies heavily on age (28.2% importance weight), which can make predictions especially sensitive to specific age boundaries.
- **Split Sensitivity:** Decisions are bound to strict threshold boundaries on variables like ap_hi and BMI established during training.

4. Final Conclusion

❖ Key Insights:

- Systolic blood pressure (ap_hi), age, and BMI are the primary CVD risk drivers. Dropped weight due to severe collinearity with BMI ($r = 0.85$), but kept both blood pressure metrics (ap_hi/ap_lo) for clinical accuracy. Dataset size (68,646 rows) was fully sufficient.

❖ **Model Optimization:**

- Constraining the baseline Decision Tree to max_depth = 7 fixed severe training overfitting. Shifting decision cutoffs below 0.50 based on sensitivity-specificity curves significantly boosted screening sensitivity across both models.

❖ **Final Selection:**

- Linear SVC (0.42 cutoff) is selected over the Tuned Decision Tree (0.465 cutoff). While both achieve 0.72 test accuracy, Linear SVC delivers higher test recall (0.73 vs 0.72), minimizing dangerous missed CVD cases (False Negatives).